

Chapter 3. **GROUPS**

3.3 Cyclic Groups

3.3.2 The Fundamental Theorem of Finite Cyclic Groups



Today we will provide a precise description of all the subgroups of a finite cyclic group.

3.3.2 The Fundamental Theorem of Finite cyclic groups

The *Fundamental Theorem of Finite Cycle Groups*, gives a **precise description of all subgroups of a finite cyclic group G** . It tells not only how many subgroups G has but also how to obtain them. To be more precise: it says that G has a unique cyclic subgroup of order k for every divisor k of n and those are the only subgroups of G .

THEOREM 3.30. FUNDAMENTAL THEOREM OF FINITE CYCLIC GROUPS.

Let $G = \langle g \rangle$ be a cyclic group of order n .

- (1) If H is a subgroup of G , then $H = \langle g^d \rangle$ for some $d|n$.
- (2) If H is any subgroup of G with order $|H| = k$, then $k|n$.
- (3) If $k|n$, then $\langle g^{n/k} \rangle$ is the unique subgroup of G of order k .

Another interpretation of the *Fundamental Theorem of Finite Cyclic Groups*

the subgroups of G are in one-to-one correspondence with the positive divisors of n

LEMMA 3.31. Let g be an element of a group G of order n . If $d|n$ then the order of the element g^d is n/d .

Proof. Let $K = \frac{n}{d}$. To show that the order of g^d is K , we need to check the following:

① $(g^d)^K \stackrel{?}{=} 1$

② If $r \geq 1$ such that $(g^d)^r = 1$, then $K|r$.

Proof of ①: $(g^d)^K = g^{dK} = g^n = 1$

because we are assuming that the order of g is n .

Proof of ②: Suppose that $r \geq 1$ such

that $(g^d)^r = 1$. Then

$1 = (g^d)^r = g^{dr}$, and since we are assuming that the order of g is n , we have that $n \mid dr$. This implies that $dr = nq$, for some integer q .

$$dr = nq = (kd)q \Rightarrow d(r - kq) = 0$$

$$\Rightarrow r - kq = 0 \Rightarrow r = kq \Rightarrow \underline{k \mid r},$$

$\uparrow$
 $d \geq 1$

as requested.

